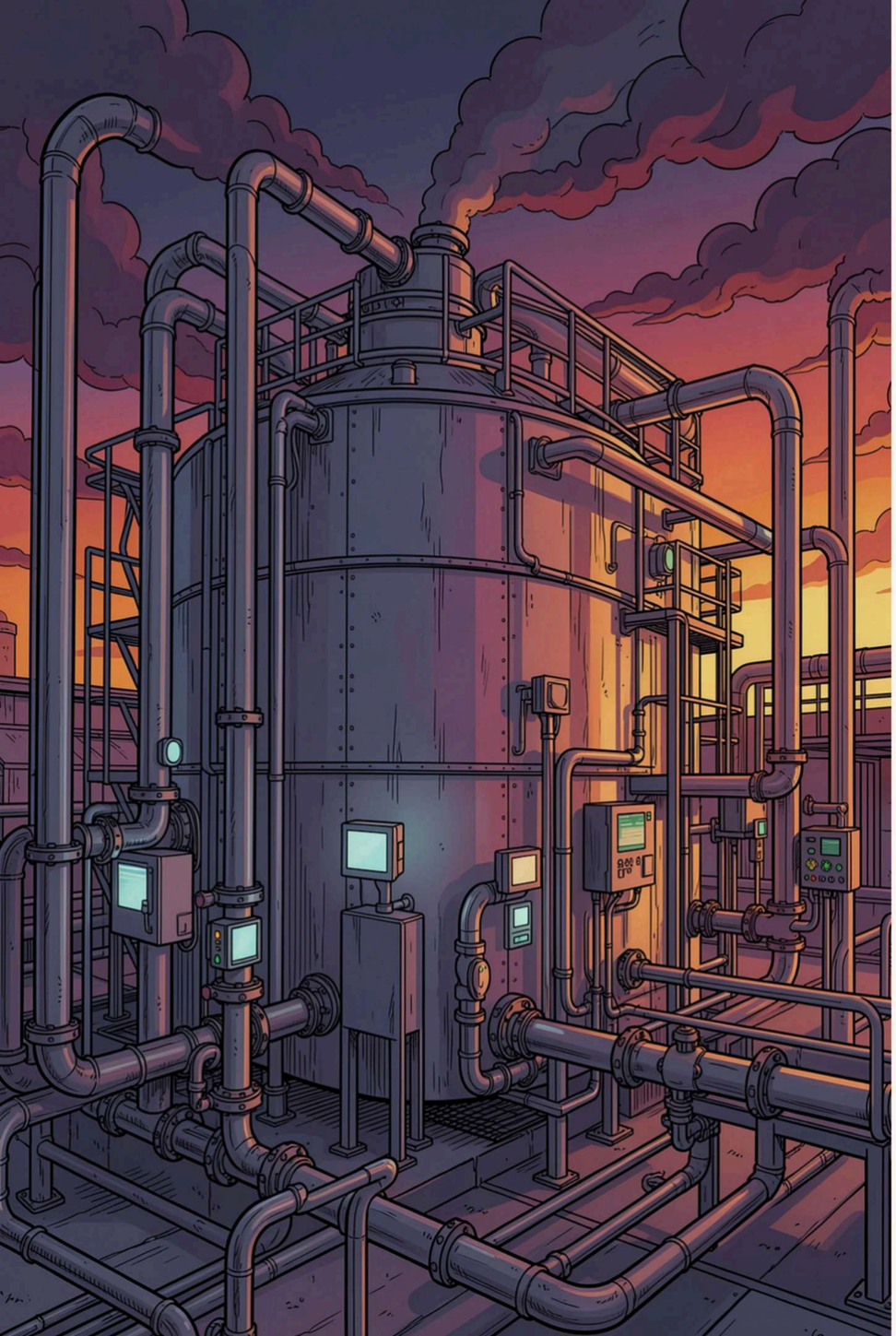


# Smart Water Tank Monitoring & Automatic Pump Control

An IoT-based system for real-time level sensing, automated pump control, and cloud-connected water management


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# Challenges & Motivation

## The Problem

### Water Overflow

Manual monitoring leads to wastage and property damage

### Energy Wastage

Inefficient pump operation inflates electricity consumption

### No Real-Time Data

Zero visibility into usage patterns or tank status

## Our Solution — 5 Implementation Tasks

1. **Sensor Integration:** Ultrasonic sensor + ESP32 + temperature sensor + TDS sensor + Flow meter for continuous acquisition.
2. **Cloud Connectivity:** Wi-Fi to ThingSpeak/Arduino Cloud for real-time transmission
3. **Automated Control:** Pump ON below 20%, OFF above 90% with critical alerts
4. **Dashboard Development:** Mobile/web interface for live monitoring.
5. **Notification System:** Push alerts for low, high, and critical conditions

# Research-Backed Solid Reasons

This project addresses a **legitimate industrial control problem** — not merely home automation. Three bodies of peer-reviewed evidence establish the engineering necessity.

1

## Industrial Process Control — Journal of Process Control

Liquid level control is a **critical process-control problem** across industries. Liquid inventories in reactors, decanters, extractors, distillation columns, and surge tanks must remain within safe operating limits — because **liquid level is generally not self-regulating**. The core engineering challenge: automatically maintain level within a predefined safe range while controlling pumps, preventing overflow/underflow, and eliminating manual intervention.

2

## Wastewater Treatment Plants — IEEE Research

The IEEE paper *"Design and Implementation of a Water Level Controller in Pumping Chamber which is Part of a Wastewater Treatment Plant"* demonstrates automatic level control in real wastewater-treatment pumping stations. Pumping chambers must be maintained within operating limits: if level becomes too high, overflow occurs — making **automatic sensing and pump control a regulatory and operational requirement**, not an option. Failure risks backup of untreated wastewater and environmental non-compliance.

3

## Chemical & Oil Industries — Journal of Loss Prevention

The *Journal of Loss Prevention in the Process Industries* examined overfill incidents in natural gas processing, oil refining, and petroleum storage. Loss of level control leads to high-level conditions and **potential loss of containment**. Unlike domestic overflow (wastage), chemical/oil tank overflow means **hazardous spill → environmental contamination → safety catastrophe**. Historical incidents at **Longford (Australia)**, **Texas City (USA)**, and **Buncefield (UK)** show how high-level conditions propagate into downstream equipment not designed to receive excess liquid — causing catastrophic failures. Automatic level monitoring is a **safety requirement**.

⚠ Overfill incidents at Longford, Texas City, and Buncefield confirm: level control failure is not an inconvenience — it is a catastrophic safety risk.

# Smart Water Tank Level Monitoring & Automatic Pump Control – IoT System Block Diagram

## Sensors / Inputs (From Water Tank)



**HC-SR04**
Ultrasonic Sensor
(Water Level)

Water Level Distance



**DS18B20**
Temperature Sensor
(Water Temperature)

Temperature Data



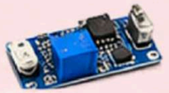
TDS Sensor
(Water Purity)

TDS Value (Data)

## Power Supply (for Pump & Relay Side Only)

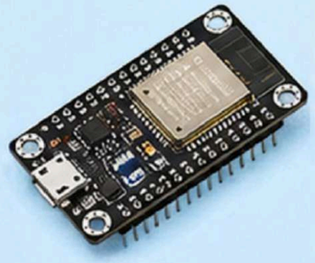


Battery
9V / 12V



DC-DC Converter
(12V/9V → 5V)

5V for Relay & Pump



## ESP32
(Data Processing, Control & IoT)

Control Signal (5V)

Alarm Signal (5V)

## Actuators & Control (Pump Side)



5V 1 Channel Relay
(Pump Control)

6V DC (Switched)



6V DC Water Pump
(Fills Tank)

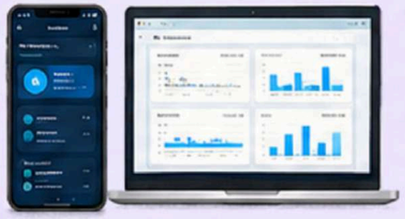


Buzzer
(Alarm)



Water Tank
(Stores Water)

## Mobile / Web Dashboard



- Water Level
- Temperature
- TDS (Water Quality)
- Flow Rate
- Total Consumption
- Alerts / Notifications



Cloud Server
(Real-time Data Storage & Analytics)

Flow Pulses
(Via ESP32)



YFS201 Flow Meter
(Measures Flow Rate & Total Consumption)

## Process Flow

1. Sensors Read Data (HC-SR04, DS18B20, TDS)
2. ESP32 Processes Data and Decides Actions
3. Controls Relay & Pump (if needed)
4. Shares Data to Cloud (via Wi-Fi)
5. User Monitors on Mobile/Web Dashboard
6. Alerts Triggered (if thresholds crossed)

# 5. Hardware and Circuit Design

The pin-level schematic below is based on the uploaded firmware, the supplied ESP32 DevKit pinout image, and the confirmed correction that the TDS analog output is connected to GPIO 35. It is a connection schematic rather than a PCB layout.

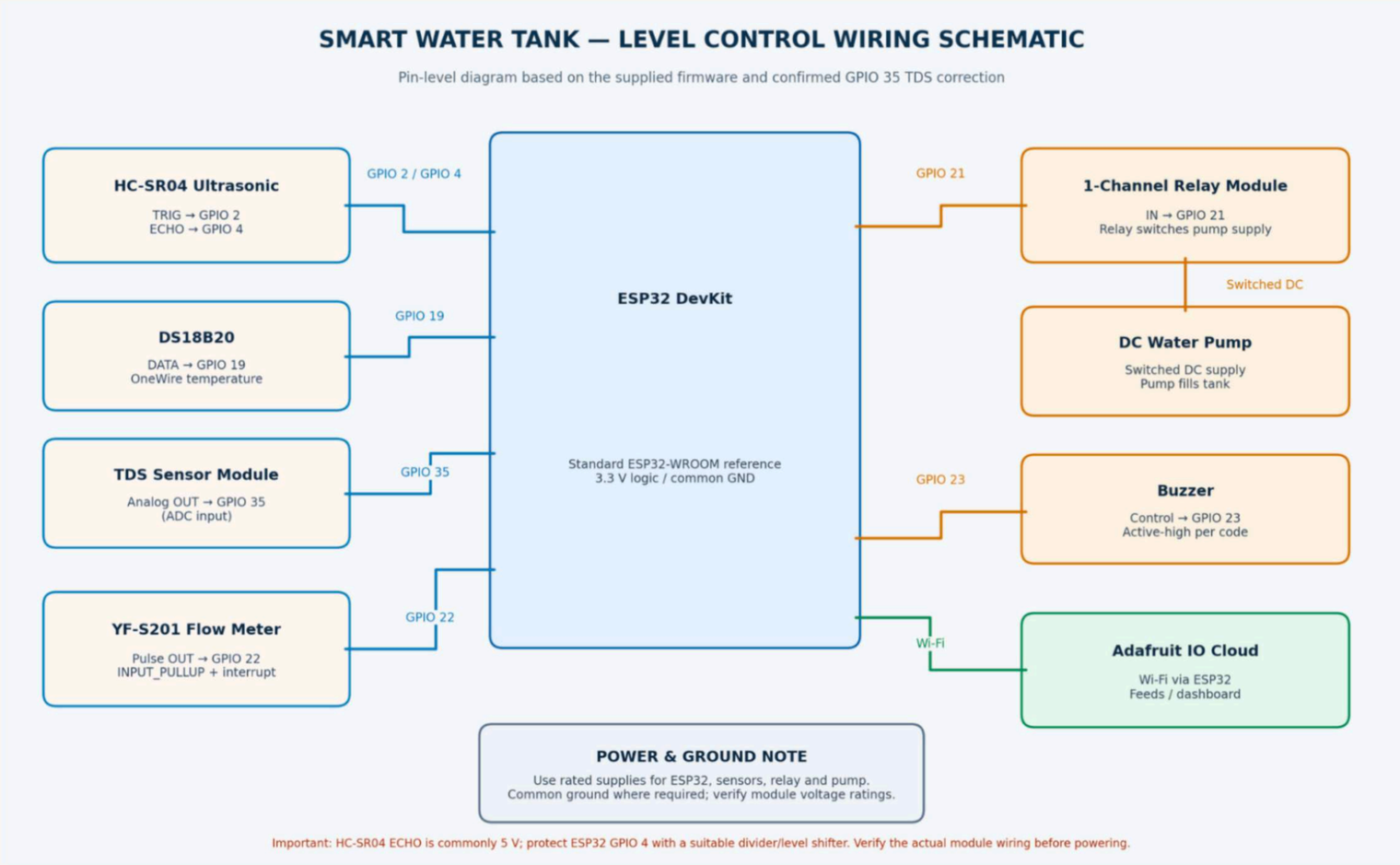
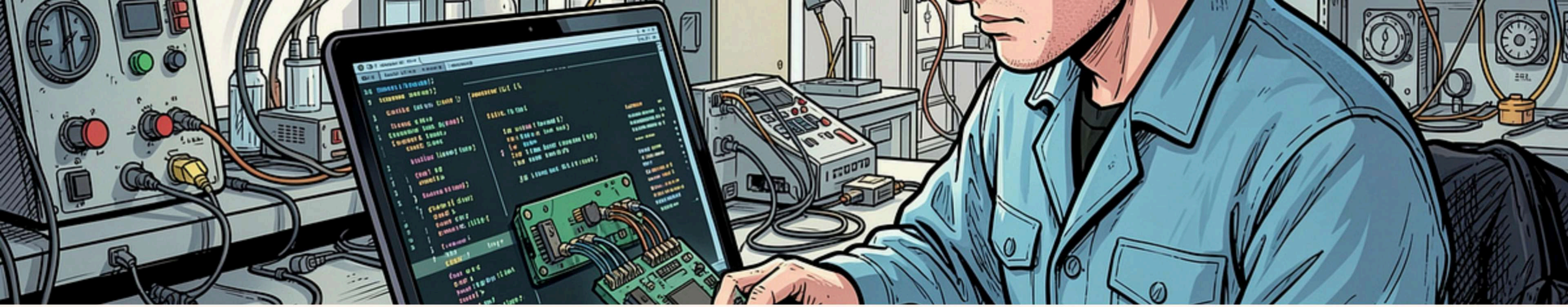



*Figure 1. ESP32-based level-control wiring schematic derived from the supplied pin configuration.*

# GPIO Mapping

| Signal / module | ESP32 pin | Notes |
| --- | --- | --- |
| Ultrasonic TRIG | GPIO 2 | Output trigger pulse. |
| Ultrasonic ECHO | GPIO 4 | Input pulse duration; protect input if sensor ECHO is 5 V. |
| Relay input | GPIO 21 | Relay active-low setting in supplied code. |
| Buzzer control | GPIO 23 | Active-high setting in supplied code. |
| Flow meter pulse | GPIO 22 | Interrupt input with INPUT_PULLUP. |
| DS18B20 data | GPIO 19 | OneWire temperature interface; physical pull-up resistor is normally required. |
| TDS analog output | GPIO 35 | Confirmed correction from GPIO 18; GPIO 35 supports ADC input. |

Power wiring must follow the ratings printed on the actual modules. The pump should be powered through a suitably rated relay and supply, not directly from an ESP32 GPIO. Grounds must be referenced appropriately for the sensor/control circuits. The HC-SR04 ECHO output is commonly 5 V, while ESP32 GPIO inputs are not 5-V tolerant; a suitable divider or level shifter is required unless the actual sensor interface already provides compatible logic levels.



# Implementation Details

## Step-by-Step Execution (Components and Communication)

01

### Hardware Setup

Mount ultrasonic sensor inside tank; connect relay to pump; integrate Temp. and TDS sensors for liquid monitoring.

02

### ESP32 Programming

Develop firmware in Arduino IDE (C++) to read sensors, process thresholds, and communicate via MQTT/HTTP.

03

### Cloud Configuration

Connect ESP32 to Adafruit.io cloud and display live parameters.



# Cloud & Mobile Interface

## Cloud Platform

Adafruit.io cloud provides remote data processing and access.

## Real-Time Dashboard

Live water-level percentage display with graphical visualization on web and mobile.

## Notification System

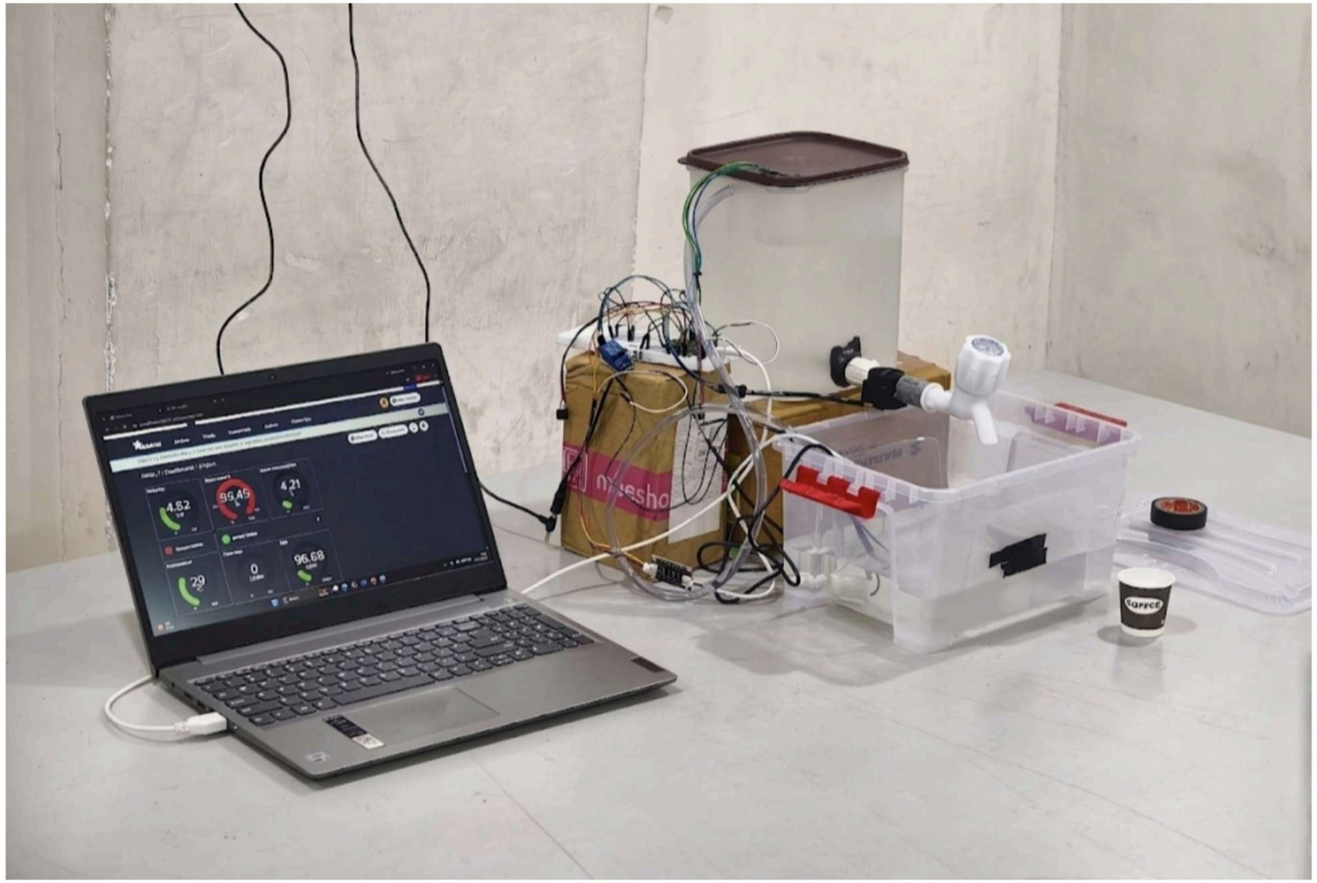
Push notifications reports triggered automatically for low-level, high-level, and critical conditions.

## Water Consumption Measurement

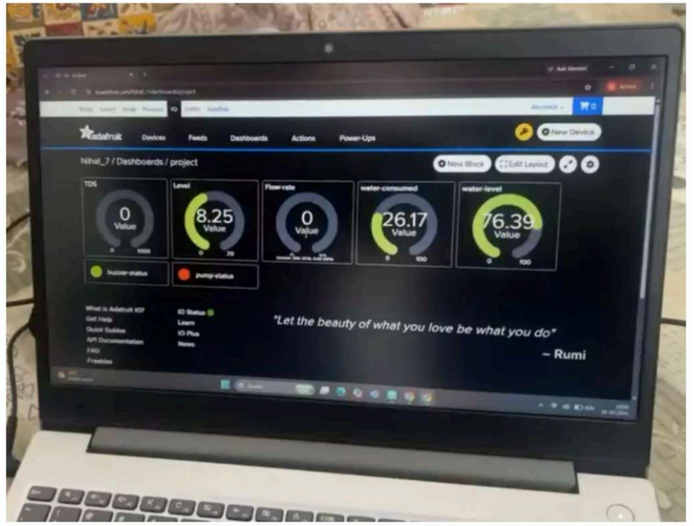
Water consumption for proactive resource management.

# Hardware Implementation & Cloud Output

Photographic evidence of the assembled prototype and connected dashboard



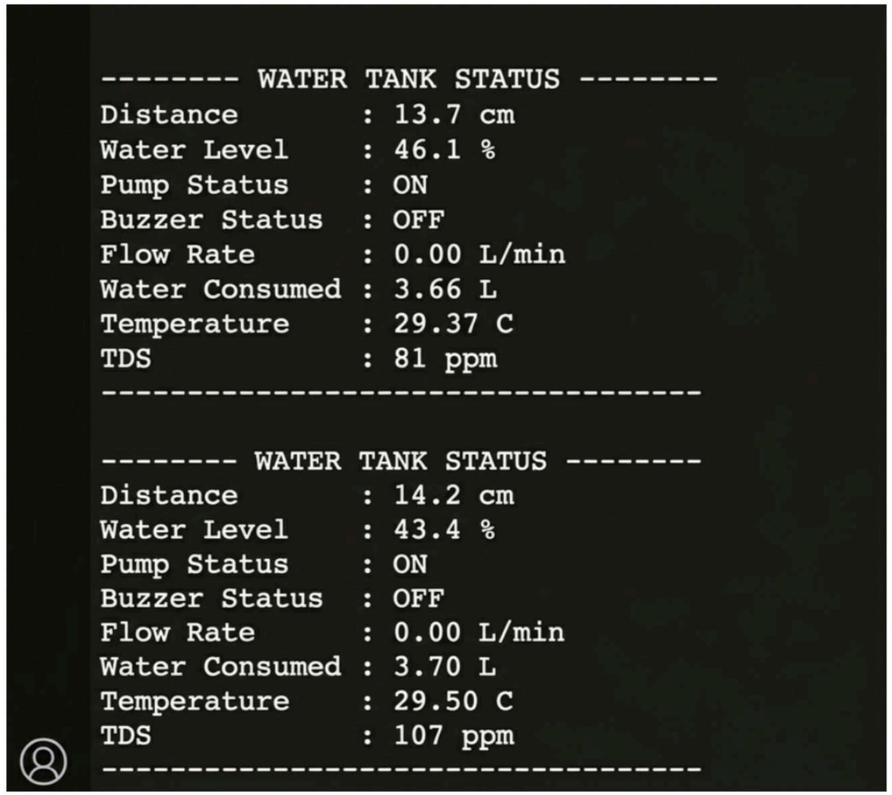
**Physical prototype setup**




**Adafruit IO dashboard**

The supplied dashboard capture shows gauge widgets for TDS, level, flow rate, water consumed, and water level, together with buzzer and pump status indicators.

# Simulation Measurements




## Recorded Hardware Readings

| Parameter | Reading 1 | Reading 2 |
|---|---|---|
| Distance | 13.7 cm | 14.2 cm |
| Water level | 46.1 % | 43.4 % |
| Pump status | ON | ON |
| Buzzer status | OFF | OFF |
| Flow rate | 0.00 L/min | 0.00 L/min |
| Water consumed | 3.66 L | 3.70 L |
| Temperature | 29.37 C | 29.50 C |
| TDS | 81 ppm | 107 ppm |

### Observation & analysis

Across the two snapshots, distance increased from 13.7 cm to 14.2 cm while water level decreased from 46.1% to 43.4%, consistent with a falling tank level. The pump remained ON and the buzzer remained OFF in both readings. Water consumed increased from 3.66 L to 3.70 L. Temperature and TDS also changed. Flow rate is displayed as 0.00 L/min in both snapshots.

*The dashboard photograph is included on the following page; unclear small on-screen values are not transcribed.*

# Conclusion

### ⚡ Efficiency

Automated monitoring eliminates manual intervention and reduces operational overhead entirely.

### 💧 Sustainability

Real-time control prevents water wastage and actively promotes conservation at scale.

### 💰 Cost Savings

Reduces electricity consumption, and minimizes water loss.

### 📈 Scalability

Architecture supports single-tank homes through to large-scale multi-tank industrial installations.

### 🤖 Future Enhancements

Water demand details, water quality analysis, and multi-tank management.

### 🌍 Impact

Applicable across residential, institutional, and industrial sectors for sustainable water resource management.

# Appendix A: Firmware Configuration and Credentials

The following values are reproduced from the supplied source-code PDF at the team's request for the offline printed report. Keep this document physically secure and do not upload or distribute it publicly.

| Configuration item | Value in supplied source |
|---|---|
| Adafruit IO username | Nihal_7 |
| Adafruit IO key | aio_gfxW34GBrLLB4PSNujVlu56N3RZY |
| Wi-Fi SSID | code |
| Wi-Fi password | code1234 |
| Ultrasonic TRIG / ECHO | GPIO 2 / GPIO 4 |
| Relay / buzzer | GPIO 21 / GPIO 23 |
| Flow / temperature | GPIO 22 / GPIO 19 |
| TDS pin | GPIO 35 (corrected by team; supplied listing shows GPIO 18) |
| Empty / full distance calibration | 22.0 cm / 4.0 cm |
| Flow pulses per litre | 450 |

Credential note: these values are sensitive even when intended for offline use. If the source credentials have been shared beyond the project team, rotate the Wi-Fi password and Adafruit IO key.

# Research References

The following peer-reviewed sources underpin the design decisions, industrial justification, and technical implementation of this project.


### Smart IoT-Based Water Tank Monitoring using ESP32 and Mobile Application

*International Journal of Machine Intelligence and Computing* — **jurcon.ums.edu.my**

### IoT-Based Industrial Wastewater Monitoring System using ESP32 and Blynk

*Journal of Engineering Technology and Applied Physics* — **journals.mmupress.com**

### Development and Implementation of a Smart Water Metering and Monitoring System for Homes with Intermittent Water Supply

*MDPI Technologies, Vol. 14(2), 135* — **mdpi.com**